

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Define Problem Statements

Online Complaint Registration System — MERN Stack Complaint
Management Platform

Phase: Brainstorming & Ideation Phase

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Classification: Project Documentation — For Academic Submission

Define Problem Statements

1. Introduction

This document formalizes the problem statements identified during the Brainstorming and Ideation phase of the Online Complaint Registration System project. Each problem statement follows the structured customer-centric framework: *I am* → *I'm trying to* → *But* → *Because* → *Which makes me feel*.

This approach helps understand the challenges faced by various stakeholders and ensures that the proposed solution directly addresses their needs. Root-cause analysis is subsequently performed to identify the underlying factors responsible for these challenges and map them to appropriate solution features.

The Online Complaint Registration System focuses on improving transparency, communication, complaint tracking, and administrative efficiency through a centralized MERN-based platform.

2. Problem Statements

PS-1: The Citizen / Complaint User

Field	Description
I am	A citizen who wants to report public service issues and track their resolution status.
I'm trying to	Register complaints easily, monitor complaint progress, and communicate with responsible authorities regarding resolution updates.
But	Existing complaint handling methods are fragmented, time-consuming, and provide little visibility into complaint status.
Because	Most traditional systems rely on manual processes, lack centralized tracking, and do not provide direct communication channels between citizens and responsible personnel.
Which makes me feel	Frustrated, ignored, and uncertain whether my complaint is being addressed.

Impact Severity:  High — Lack of transparency reduces citizen trust and delays issue resolution.

PS-2: The Agent / Complaint Handler

Field	Description
I am	A field agent responsible for resolving assigned complaints efficiently.
I'm trying to	View assigned complaints, update complaint status, communicate with citizens, and complete resolutions within expected timelines.
But	Complaint information is often scattered across multiple communication channels, making coordination difficult.
Because	Traditional systems lack centralized assignment workflows, real-time communication tools, and proper status tracking mechanisms.
Which makes me feel	Overwhelmed, less productive, and unable to provide timely responses to citizens.

Impact Severity:  High — Inefficient workflows lead to delayed complaint resolution and reduced service quality.

PS-3: The Administrator

Field	Description
I am	An administrator responsible for managing complaints, users, agents, and overall platform operations.
I'm trying to	Monitor complaint statistics, approve agents, assign complaints, and ensure service efficiency.
But	Existing systems provide limited visibility into complaint trends and agent performance.
Because	There is no centralized dashboard capable of aggregating complaint data, monitoring assignments, and generating actionable insights.
Which makes me feel	Concerned about operational inefficiency and unable to make informed management decisions.

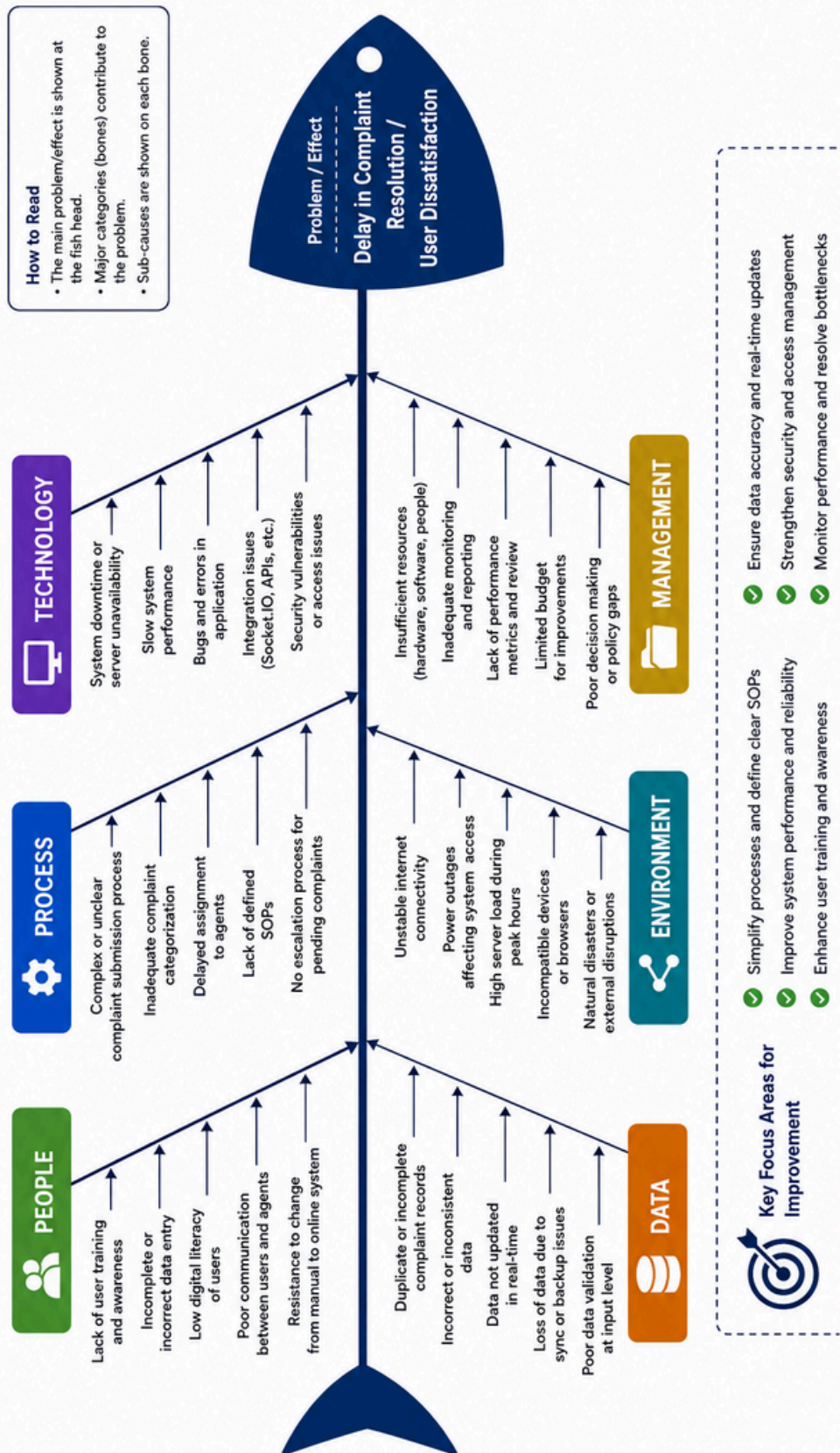
Impact Severity:  High — Poor administrative oversight negatively impacts organizational effectiveness and citizen satisfaction.

3. Root-Cause Analysis (Ishikawa Diagram)

To provide clear documentation boundary guidelines, a precise half-page structural layout allocation has been formatted below for the integration of the Root-Cause Analysis visualization model.

FISHBONE (ISHIKAWA) ROOT CAUSE DIAGRAM

Online Complaint Registration System



4. Problem-to-Feature Mapping

Root Cause	Proposed Solution Feature
Manual complaint registration	Online complaint registration portal
Lack of complaint tracking	Real-time complaint tracking dashboard
Delayed communication	Socket.IO real-time chat system
No notification mechanism	Real-time notification service
Weak authentication	JWT Authentication System
Poor access control	Role-Based Access Control (Citizen / Agent / Admin)
Lack of centralized data	MongoDB Atlas centralized database
Difficult complaint assignment	Admin complaint assignment workflow
Limited administrative visibility	Analytics Dashboard
No feedback collection	Feedback & Rating System
Poor scalability	MERN architecture with cloud deployment
Lack of status monitoring	Complaint lifecycle management system

5. Conclusion

The identified problem statements represent the three primary stakeholders of the Online Complaint Registration System: citizens, agents, and administrators. The analysis highlights significant challenges related to complaint registration, tracking, communication, assignment, and management.

The root-cause analysis reveals that inefficiencies primarily stem from fragmented processes, lack of centralized communication, inadequate monitoring capabilities, and insufficient technological support. To address these issues, the proposed system introduces secure authentication, role-based access control, complaint management workflows, real-time communication through Socket.IO, notifications, analytics dashboards, and feedback mechanisms.

All identified root causes have corresponding solution features that directly contribute to the design and implementation of the Online Complaint Registration System.